

COMPANY BRAIN

The Institutional Memory Engine for Modern Organizations

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"Organizations don't have a data problem. They have a memory problem."

1. EXECUTIVE SUMMARY

Company Brain is an AI-powered institutional memory layer engineered for modern, knowledge-intensive organizations. At its core, Company Brain addresses a structural challenge that has plagued enterprises for decades: the systematic fragmentation, decay, and loss of organizational intelligence. As companies scale and teams become distributed, critical knowledge — why decisions were made, how problems were solved, what was tried before — becomes buried across dozens of disconnected tools, locked inside the minds of individual employees, or simply lost to time.

Company Brain solves this by acting as the connective tissue across an organization's digital ecosystem. It ingests, indexes, and reasons across enterprise communication platforms (Slack, Gmail), knowledge repositories (Notion, Google Drive), engineering systems (GitHub, Linear), and internal documentation. The result is a secure, permission-aware, citation-backed AI system capable of answering the hardest questions organizations ask: not just "what do we know," but "why do we know it, when did we learn it, and how confident should we be today?"

Why It Matters

The modern enterprise generates an unprecedented volume of information daily. Yet the majority of this information is never effectively retrieved. According to McKinsey, knowledge workers spend an average of 1.8 hours per day searching for information — approximately 9.3 hours per week per employee. For a company of 500 employees, this represents over 4,600 hours of lost productivity every single week. At an average fully-loaded cost of \$75/hour, this translates to a staggering \$18 million in annual productivity loss per organization.

The problem is not storage. The problem is that existing storage is semantically inert. Files sit in Drive. Messages decay in Slack. Decisions evaporate when employees leave. No existing enterprise tool reconstructs the organizational reasoning behind those artifacts — the context, the alternatives considered, the rationale for the chosen path.

The Market Gap

Enterprise search tools surface documents. AI copilots generate responses. Knowledge management platforms organize wikis. But none of these systems address the fundamental gap: the ability to reconstruct institutional reasoning across time, across tools, and across people — while respecting the access permissions that govern every enterprise environment. Company Brain fills this gap with a purpose-built architecture that combines temporal intelligence, cross-platform memory synthesis, permission-aware retrieval, and knowledge decay detection.

Why Now

Three macro forces converge to make this the ideal moment for Company Brain. First, the proliferation of AI-native tooling has created an expectation among enterprise users that AI should be deeply integrated into their work. Second, the rise of distributed and asynchronous work has dramatically accelerated the rate at which institutional knowledge is fragmented across tools and time zones. Third, advances in large language model capabilities — particularly in retrieval-augmented generation and tool use — now make it technically feasible to build a system of this sophistication at hackathon pace.

Strategic Importance

Company Brain is not a productivity feature. It is infrastructure. Much like how cloud storage became the default substrate for enterprise data, institutional memory infrastructure will become the default substrate for enterprise reasoning. Organizations that deploy Company Brain gain a compounding intelligence advantage: the longer it runs, the richer its memory, and the more powerful its retrieval. This document defines Phase 0 — the discovery and strategic definition that precedes all architecture, engineering, and deployment decisions.

2. PROBLEM STATEMENT

2.1 The Enterprise Knowledge Crisis

Modern enterprises operate in a paradox: they have never generated more information, and they have never been less able to make use of it. Every Slack message, every GitHub pull request comment, every Google Doc draft, every email thread represents a fragment of institutional intelligence. Yet these fragments are almost never systematically preserved, indexed, or made retrievable in a way that supports organizational decision-making.

This crisis manifests across six interconnected dimensions, each compounding the others.

2.1.1 Fragmented Information Silos

The average enterprise deploys 254 software applications (Okta, Business at Work 2023). Each application creates its own data island — Slack messages live in Slack, GitHub issues live in GitHub, Notion documents live in Notion. There is no semantic bridge connecting these islands. When an engineer needs to understand why a particular architectural decision was made 18 months ago, they must manually search across Slack, GitHub, Confluence, JIRA, email, and potentially interviews with colleagues who were present. This process is slow, incomplete, and heavily dependent on who happens to be available.

Enterprise Example: A fintech company decides to deprecate a microservice. The decision was documented in a Confluence page, discussed in a Slack thread, referenced in a GitHub PR description, and mentioned in a Linear ticket. Six months later, a new engineer proposes rebuilding the same microservice. Without an institutional memory layer, they have no efficient path to discover the prior decision or the reasoning behind it.

2.1.2 Institutional Knowledge Loss

When an employee leaves an organization, they take with them a disproportionate share of institutional knowledge. According to Deloitte, organizations lose between 30% and 85% of role-specific knowledge when a tenured employee departs. This knowledge loss is not limited to explicit documentation. It includes: the reasoning behind undocumented decisions, the informal relationships that enabled cross-team coordination, the historical context that makes current systems intelligible, and the tacit understanding of why certain approaches were abandoned.

Enterprise Example: A senior product manager at a SaaS company resigns after four years. They were the primary architect of the company's pricing strategy. Within six months, the incoming PM team makes three pricing decisions that directly contradict strategic choices made — and extensively debated — under their predecessor. The institutional cost is hundreds of thousands of dollars in revenue adjustment and months of strategic confusion.

2.1.3 Operational Inefficiency

Beyond the cost of lost knowledge, the ongoing inefficiency of knowledge retrieval imposes a continuous operational tax on every enterprise. Research consistently demonstrates that knowledge workers spend between 20% and 35% of their working hours searching for information. This is not idle browsing — it is purposeful, necessary work that is being performed at dramatically below optimal efficiency due to the absence of a unified, semantically intelligent retrieval layer.

2.1.4 Decision Opacity

Perhaps the most strategically damaging manifestation of the knowledge crisis is decision opacity: the inability to reconstruct the reasoning behind past organizational choices. When a company's board asks why a particular market was entered or exited, when a regulatory body asks for documentation of a compliance decision, or when a due diligence team asks for the rationale behind a product architecture — organizations frequently cannot provide coherent, traceable answers. This opacity creates legal, reputational, and strategic risk.

2.1.5 Employee Turnover and Knowledge Decay

The U.S. Bureau of Labor Statistics reports average annual employee turnover rates of 22% across professional services sectors. In high-growth tech companies, this rate frequently exceeds 30%. Each departure represents a knowledge loss event. Over a five-year window, most

companies experience near-total turnover of their original teams, yet their institutional knowledge systems are not designed to preserve continuity across these transitions. The result is organizational amnesia at scale.

2.1.6 Onboarding Inefficiencies

The average enterprise onboarding process takes between 3 and 12 months for a knowledge worker to reach full productivity, with a significant portion of that time spent learning institutional context that is not formally documented anywhere. New employees must shadow colleagues, attend redundant meetings, and accumulate organizational knowledge through time-consuming osmosis. An institutional memory layer that can answer questions like "how does our pricing team think about enterprise deals" or "what was the context behind our API rate limiting decision" would dramatically compress this ramp time.

3. ROOT CAUSE ANALYSIS

The enterprise knowledge crisis is not an accident. It is the predictable outcome of a specific set of technological, organizational, and cultural failures. Understanding these root causes is essential to designing a solution that addresses the problem at a structural level rather than applying superficial patches.

3.1 Technological Fragmentation

The proliferation of SaaS applications has created a structurally fragmented technological landscape. Each tool optimizes for its own surface area — Slack for real-time communication, Notion for structured documentation, GitHub for code collaboration — but no tool is designed to serve as the connective tissue between these systems. The result is what Gartner describes as "application sprawl": an environment where the total information surface is larger than any individual or system can navigate effectively.

3.2 Human Dependency on Tribal Knowledge

Most enterprise knowledge systems are not systems at all — they are people. The engineer who has been at the company for seven years, the product manager who remembers the original market analysis, the sales leader who knows which customer segments were tried and abandoned — these individuals serve as living institutional memory repositories. This is a fragile, non-scalable, and ultimately unsustainable architecture. When these individuals leave, are unavailable, or simply forget, the institutional knowledge they represent is gone.

3.3 Tool Sprawl Without Semantic Integration

Even when individual tools provide good internal search capabilities, there is no cross-tool semantic layer. A Slack search finds Slack messages. A GitHub search finds GitHub content. No existing tool answers the question: "across all of our communication and documentation systems, what does our organization know about X, and how has that knowledge evolved over time?" This semantic integration gap is the core architectural failure that Company Brain addresses.

3.4 Poor Institutional Memory Preservation

Institutional knowledge is rarely preserved intentionally. Documentation is created for current use, not for historical retrieval. Meeting notes are taken inconsistently. Decision rationale is seldom captured at all. The cultural norm in most organizations is to create information for immediate consumption, not for long-term organizational memory. The absence of a memory-first culture means that valuable institutional intelligence evaporates as a matter of routine.

3.5 Lack of Temporal Reasoning in Enterprise Systems

Existing enterprise search and AI systems treat information as static. A document from three years ago is retrieved with the same confidence as a document from last week, regardless of whether the domain has changed dramatically in the interim. No existing system reasons about the temporal confidence of retrieved information — whether knowledge is fresh, stale, superseded, or potentially contradicted by more recent developments. This temporal blindness leads to the retrieval of outdated information with inappropriate confidence.

3.6 Root Cause Summary Diagram

Root Cause	Primary Effect	Downstream Impact
Tool Fragmentation	Siloed information islands	Manual, incomplete retrieval
Tribal Knowledge Dependency	Knowledge locked in individuals	High turnover knowledge loss
No Semantic Integration	Cross-tool queries impossible	Redundant work, poor decisions
Poor Memory Preservation	Decisions undocumented	Decision opacity, legal risk
No Temporal Reasoning	Stale info retrieved confidently	Outdated decisions, rework

4. MARKET OPPORTUNITY

4.1 TAM / SAM / SOM Analysis

The market opportunity for Company Brain sits at the intersection of three large and rapidly growing categories: Enterprise Knowledge Management, AI-Powered Enterprise Search, and AI-

Native Infrastructure. Each of these categories is experiencing structural tailwinds driven by the accelerating adoption of AI across enterprise environments.

Total Addressable Market (TAM)

The global enterprise knowledge management market was valued at approximately \$395 billion in 2023 and is projected to reach \$1.1 trillion by 2030, growing at a CAGR of 16.3% (Grand View Research). This market encompasses all software, services, and infrastructure designed to help organizations capture, manage, and leverage their institutional knowledge. Company Brain addresses this market at its highest-value layer: AI-powered, cross-platform, temporally intelligent memory infrastructure.

Serviceable Addressable Market (SAM)

Within the TAM, the SAM for Company Brain is defined by organizations that: (1) employ more than 100 knowledge workers, (2) use three or more of the target enterprise platforms (Slack, GitHub, Notion, Google Drive, etc.), and (3) operate in sectors with high knowledge intensity and turnover (technology, professional services, financial services, life sciences). This segment represents approximately 850,000 organizations globally, with an estimated SAM of \$180 billion by 2027.

Serviceable Obtainable Market (SOM)

For the initial three-year commercial horizon, Company Brain targets mid-market and enterprise technology companies (100–10,000 employees) in North America and Western Europe. Assuming a market penetration rate of 0.5% of SAM in year three, at an average annual contract value of \$45,000, the three-year SOM is approximately \$380 million. This is a conservative estimate that does not account for platform expansion, usage-based pricing upside, or adjacent market opportunities.

Metric	Value	Basis	Time Horizon
TAM	\$1.1 Trillion	Global enterprise KM market	2030
SAM	\$180 Billion	Target enterprise segment	2027
SOM	\$380 Million	0.5% SAM penetration @ \$45K ACV	Year 3

4.2 Growth Catalysts

- Enterprise AI adoption accelerating: Gartner predicts that 80% of enterprise software will embed AI capabilities by 2026, creating demand for AI-native memory infrastructure.
- Distributed work normalization: Post-pandemic, knowledge fragmentation across time zones and tools has become a structural enterprise challenge, not a temporary anomaly.

- Regulatory pressure: Increasing regulatory requirements around decision documentation (SOX, GDPR, AI Act) create compliance-driven demand for auditable institutional memory systems.
- Talent market volatility: Continued high turnover in technology sectors creates persistent demand for turnover-resilient knowledge infrastructure.
- AI-native organization emergence: A new generation of companies is being built AI-first, creating demand for memory infrastructure from day one rather than as an afterthought.

5. PROJECT VISION

5.1 Vision Statement

Company Brain becomes the Memory Infrastructure Layer for AI-Native Organizations — the substrate through which enterprises think, remember, and reason at institutional scale.

This is not a vision of incremental improvement to enterprise search. It is a vision of categorical transformation in how organizations relate to their own intelligence. Just as cloud infrastructure abstracted away the complexity of managing physical servers, Company Brain abstracts away the complexity of managing institutional knowledge — making it universally accessible, temporally aware, permission-safe, and continuously improving.

5.2 Five-Year Vision Roadmap

Year	Milestone	Description
Year 1	Foundation	Core connectors for Slack, GitHub, Notion, Google Drive. Permission-aware RAG engine. Citation-backed retrieval. Initial enterprise pilots.
Year 2	Intelligence	Temporal reasoning engine. Knowledge decay detection and scoring. Organizational context graph. Cross-platform memory synthesis. 50+ enterprise customers.
Year 3	Platform	Open connector SDK. Third-party integrations marketplace. Agent framework for automated knowledge workflows. Series A funding. 500+ customers.
Year 4	Intelligence Layer	Proactive knowledge surface: Company Brain surfaces relevant institutional memory before it is asked for. Organizational intelligence analytics dashboard.

Year 5	Infrastructure	Company Brain becomes the default memory substrate for AI-native organizations globally. Embedded in enterprise AI stacks as foundational infrastructure.
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6. TARGET USER PERSONAS

6.1 Engineering Teams

- Pain Points: Unable to retrieve architectural decision history; rediscover solved problems; onboard to legacy codebases; understand why deprecated patterns exist.
- Current Workflow: Manually search GitHub PRs, Slack channels, Confluence; ask senior engineers; read outdated wikis.
- How Company Brain Helps: Instant retrieval of architectural reasoning with citations from GitHub, Slack, and documentation; temporal confidence scoring prevents reliance on outdated decisions.

6.2 Product Managers

- Pain Points: Cannot reconstruct why product decisions were made; lose context when transitioning between roadmap cycles; struggle to maintain strategic continuity across team changes.
- Current Workflow: Maintain personal notes; rely on colleagues who were present; search Notion and Linear manually.
- How Company Brain Helps: Cross-platform retrieval of product decision history with full context; automated detection of knowledge staleness; onboarding acceleration for incoming PMs.

6.3 Executive Leadership

- Pain Points: Cannot get authoritative answers to strategic questions; board preparation requires manual aggregation across teams; due diligence processes are painful and slow.
- Current Workflow: Commission manual reports; hold review meetings; rely on institutional knowledge of long-tenured executives.
- How Company Brain Helps: Executive-grade query interface with citation-backed, temporally aware responses; strategic decision audit trails; regulatory documentation support.

6.4 Sales Teams

- Pain Points: Cannot access institutional knowledge about past customer conversations, lost deals, competitor insights, or pricing decisions; repeat discovery work on known accounts.
- Current Workflow: Search CRM, email, Slack; ask account team members; rely on memory.
- How Company Brain Helps: Permission-aware retrieval of customer interaction history; competitive intelligence synthesis; institutional playbook retrieval.

6.5 New Employees

- Pain Points: 3–12 month ramp time; unable to access organizational context; dependent on colleague availability for institutional questions; high risk of repeating past mistakes.
- Current Workflow: Shadow colleagues; attend onboarding sessions; read documentation of varying quality and freshness.
- How Company Brain Helps: On-demand institutional context retrieval; "why" questions answered with citations; knowledge decay flags prevent onboarding to outdated information.

6.6 Knowledge Workers (General)

- Pain Points: 1.8 hours/day lost to information search; uncertainty about information accuracy; inability to trust retrieved knowledge without validation.
- Current Workflow: Multi-tool manual search; ask colleagues; accept uncertainty.
- How Company Brain Helps: Unified semantic search across all enterprise tools; temporal confidence scoring; citation-backed responses for verification.

7. PROJECT GOALS

7.1 Primary Goals (Hackathon Scope)

1. Build a functional, demo-ready institutional memory retrieval system that ingests data from at least three enterprise platforms.
2. Implement permission-aware retrieval that enforces enterprise access controls at query time.
3. Demonstrate temporal intelligence through knowledge decay detection and confidence scoring.
4. Deliver citation-backed retrieval that surfaces the source, author, and timestamp of every retrieved knowledge fragment.
5. Create a compelling, technically credible demo that communicates the value proposition clearly to hackathon judges.

7.2 Secondary Goals

- Implement a cross-platform knowledge synthesis layer that identifies related information fragments across different tools.
- Build an organizational context graph that maps the relationships between people, teams, decisions, and knowledge artifacts.
- Demonstrate the system's ability to answer complex, multi-source questions that no individual enterprise tool could answer alone.
- Produce documentation and architecture diagrams that communicate the system's design credibility.

7.3 Long-Term Strategic Goals

- Establish Company Brain as the institutional memory infrastructure standard for AI-native organizations.
- Build a connector ecosystem covering the top 25 enterprise platforms used by knowledge-intensive organizations.
- Develop proprietary temporal reasoning algorithms that represent a defensible technical moat.
- Create network effects through organizational context graphs that become more valuable as organizational complexity grows.
- Achieve regulatory compliance certifications (SOC 2, ISO 27001) to unlock enterprise sales channels.

8. SUCCESS CRITERIA

8.1 Technical Success Metrics

Metric	Target	Measurement Method
Query response latency	< 3 seconds	End-to-end timing from query submission to response display
Retrieval relevance accuracy	> 85%	Manual evaluation of top-5 retrieved results against query intent
Citation precision	> 90%	% of citations that correctly trace to the stated source document
Cross-platform synthesis	>= 3 sources	% of complex queries that surface results from multiple platforms
Permission enforcement accuracy	100%	Zero cross-permission retrievals in controlled test scenarios
System availability during demo	>= 99.9%	Zero unplanned downtime during judging window

8.2 Business Impact Metrics

- Search-Time Reduction: Target 70% reduction in time-to-insight for complex institutional questions versus manual search.
- Onboarding Acceleration: Demonstrate ability to answer 80%+ of typical new-hire contextual questions from institutional memory alone.
- Decision Context Recovery: Successfully reconstruct reasoning context for 3+ historical decisions in live demo scenarios.
- Knowledge Staleness Detection: Flag at least 90% of artificially aged test documents as potentially stale.

8.3 Hackathon-Specific Success Criteria

- **Demo Readiness:** A fully functional, stable demonstration that can be run live without preparation or precondition.
- **Technical Credibility:** Architecture documentation and code quality sufficient to withstand technical scrutiny from expert judges.
- **Originality Score:** Differentiation from generic RAG solutions clearly communicated through temporal intelligence, permission awareness, and organizational reasoning features.
- **Presentation Clarity:** A narrative that communicates the problem, solution, differentiation, and impact within a five-minute presentation window.

9. NON-FUNCTIONAL REQUIREMENTS

Category	Requirement	Implementation Approach
Security	End-to-end encryption for all data at rest and in transit	AES-256 encryption at rest; TLS 1.3 in transit; zero-trust architecture
Permission Integrity	Retrieval must respect source-system access permissions exactly	Permission metadata ingested at index time; enforced at query time before result delivery
Scalability	Support indexing of up to 10M documents without degradation	Vector database with horizontal sharding; async ingestion pipeline with queue-based backpressure
Reliability	99.9% uptime SLA for production deployments	Multi-region deployment; automatic failover; health monitoring with alerting
Explainability	Every response must include source citations with confidence scores	Citation extraction enforced at RAG layer; confidence scoring via temporal decay model
Observability	Full query and retrieval logging with latency telemetry	Structured logging to observability platform; distributed tracing across ingestion and retrieval paths
Performance	P95 query response time under 3 seconds	Pre-computed embeddings; approximate nearest-neighbor search; LLM response streaming
Data Residency	Customer data must not leave designated geographic region	Region-specific deployment configuration; no cross-region data replication of customer content

10. RISK ASSESSMENT

The following risk matrix identifies the primary risks associated with the Company Brain hackathon project, with probability and impact rated on a 1–5 scale (1 = Low, 5 = Critical).

Risk	Probability	Impact	Mitigation Strategy
Connector complexity & OAuth friction	4 / 5	4 / 5	Pre-build connectors before hackathon start; use mock data adapters for unstable APIs; prioritize most stable connectors (GitHub, GDrive).
Perceived as generic RAG application	3 / 5	5 / 5	Lead demo with temporal intelligence and permission-awareness features; prepare clear differentiation narrative; show features no standard RAG system can replicate.
Retrieval quality insufficient for demo	3 / 5	5 / 5	Use curated seed data that maximizes retrieval quality; tune chunking strategy pre-demo; prepare fallback queries with known-good retrieval paths.
Time constraints & scope overrun	4 / 5	3 / 5	Strict MVP scope definition in Phase 0; time-boxed feature development; demo-readiness as primary exit criterion; secondary features cut if behind schedule.
Demo environment instability	2 / 5	5 / 5	Run on stable cloud infrastructure (not local); pre-warm all services; record fallback demo video; test demo script 5+ times before judging.
Weak differentiation from Notion AI / Glean	3 / 5	4 / 5	Emphasize temporal intelligence, organizational reasoning reconstruction, and cross-platform synthesis — features Notion AI and Glean do not provide.
LLM response quality inconsistency	2 / 5	3 / 5	Use structured prompting with explicit citation instructions; test with diverse query types; implement response quality guard against hallucinated citations.

11. COMPETITIVE LANDSCAPE

Company Brain operates in a competitive landscape that includes enterprise search tools, AI copilots, traditional knowledge management platforms, and general-purpose RAG assistants. While each category addresses a subset of the problem, none provides the complete institutional memory infrastructure that Company Brain is designed to deliver.

Feature	Company Brain	Glean / Guru	Notion AI	Generic RAG
Cross-platform retrieval	Yes	Partial	No	Partial
Temporal intelligence	Yes (core feature)	No	No	No
Permission-aware retrieval	Yes (enforced)	Partial	No	No

Knowledge decay detection	Yes	No	No	No
Citation-backed responses	Yes	Partial	Partial	Variable
Organizational context graph	Yes	No	No	No
Reasoning reconstruction	Yes	No	No	No
Open connector ecosystem	Planned	Yes	No	No

The competitive analysis reveals that while tools like Glean provide broad enterprise search with reasonable connector coverage, they do not address the temporal dimension of institutional knowledge or the reasoning reconstruction challenge. Notion AI is constrained to the Notion ecosystem. Generic RAG assistants lack the permission architecture, connector breadth, and temporal intelligence required for enterprise deployment. Company Brain is uniquely positioned to occupy the institutional memory infrastructure category — a position no current competitor owns.

12. STRATEGIC DIFFERENTIATORS

12.1 Temporal Intelligence

Temporal Intelligence is Company Brain's most distinctive technical capability and its primary defensible moat. Unlike all existing enterprise retrieval systems, Company Brain does not treat information as static. Every knowledge fragment ingested by the system is associated with a temporal confidence model that decays over time based on domain-specific volatility parameters.

A pricing decision documented 18 months ago in a market that has seen three competitive entries since then should be retrieved with significantly lower confidence than a technical architecture decision that has been stable for the same period. Company Brain's temporal reasoning engine encodes this intuition algorithmically, assigning decay rates based on document type, domain classification, and observed revision frequency. The result is a retrieval system that answers not just "what does our organization know" but "how confident should we be in what we know, given when we learned it?"

12.2 Permission-Aware Retrieval

Enterprise knowledge systems that ignore permissions are not enterprise systems — they are security liabilities. Company Brain is architected with permission integrity as a first-class, non-negotiable design constraint. When content is ingested from any source system, the access permissions governing that content in the source system are ingested alongside the content itself and stored as immutable metadata in the vector store.

At query time, before any retrieval result is surfaced to the requesting user, the system enforces a permission check that verifies the requesting user's identity against the permission metadata of each candidate result. Results the user would not have been able to access in the source system are filtered before they are included in the response. This enforcement is architectural — it cannot be bypassed by prompt injection or user privilege escalation attempts.

12.3 Organizational Reasoning Reconstruction

The most sophisticated capability in Company Brain's feature set is organizational reasoning reconstruction: the ability to synthesize a coherent, contextualized account of why an organizational decision was made by drawing on fragments from multiple sources, multiple time points, and multiple contributors.

When a user asks "why did we choose PostgreSQL over MongoDB for our primary data store," Company Brain does not simply retrieve a single document. It identifies the GitHub PR discussion where the decision was first proposed, the Slack thread where the trade-offs were debated, the architecture document where the decision was formalized, and the Notion page where the decision was retrospectively evaluated. It synthesizes these fragments into a coherent account, attributes each claim to its source, and presents the reasoning with full provenance.

12.4 Knowledge Decay Detection

Knowledge Decay Detection is the system capability that prevents organizations from making decisions based on stale institutional memory. The decay detection engine monitors the freshness of indexed knowledge fragments relative to: the rate of change in the relevant domain, the presence of subsequent revisions or contradictions, and the elapsed time since the knowledge was created or last validated.

When a user queries the system for information in a domain where the indexed knowledge is potentially stale, the system surfaces a decay warning alongside the retrieved results. This warning includes the last known validation date, the detected indicators of potential staleness, and a recommendation to validate the information against current sources. This capability is critical for high-stakes enterprise decision-making environments where acting on outdated information can have significant business or regulatory consequences.

12.5 Institutional Context Graph

Underlying all of Company Brain's retrieval capabilities is an institutional context graph: a dynamic, continuously updated graph data structure that maps the relationships between knowledge artifacts, people, teams, decisions, and time. The context graph enables Company Brain to answer relational questions that vector similarity search alone cannot address: questions about who was involved in a decision, which teams were consulted, how a decision evolved over time, and what other decisions depended on a given piece of institutional knowledge.

The context graph is a long-term strategic asset that becomes more valuable as the organization's knowledge base grows. Unlike vector embeddings, which represent point-in-time semantic similarity, the context graph represents the evolving relational structure of organizational intelligence — a uniquely defensible data asset that competitors cannot replicate without equivalent deployment tenure.

13. MVP DEFINITION

13.1 Features Included in Hackathon MVP

- Multi-platform ingestion: Connectors for GitHub (PRs, issues, discussions), Slack (messages, threads, channels), Notion (pages, databases), and Google Drive (documents, spreadsheets).
- Vector embedding pipeline: Chunking, embedding generation using OpenAI text-embedding-3-large, and storage in a vector database (Pinecone or pgvector).
- Permission-aware retrieval: Role-based access control metadata ingested and enforced at query time.
- Temporal confidence scoring: Basic decay model applied to retrieved results based on document age and domain classification.
- Citation-backed responses: Every LLM response includes explicit citations to source documents with author, timestamp, and platform metadata.
- Query interface: Clean, professional web UI for submitting institutional memory queries and viewing citation-backed responses.
- Demo scenario support: Pre-loaded seed data enabling compelling demo scenarios across engineering, product, and executive use cases.

13.2 Scope Boundaries (Out of MVP)

- Real-time streaming ingestion (MVP uses batch ingestion on scheduled intervals).
- Full organizational context graph (MVP uses lightweight relationship tagging, not full graph database).
- Advanced proactive knowledge surfacing (MVP is query-driven only).
- Mobile application interface.
- Enterprise SSO and identity provider integration.
- Connector SDK for third-party integration development.

13.3 Technical Stack

Component	Technology Choice	Rationale
LLM	OpenAI GPT-4o / GPT-4-turbo	Best-in-class instruction following; native tool use; hackathon sponsor.
Embeddings	OpenAI text-embedding-3-large	High-dimensional embeddings; strong semantic retrieval performance.
Vector Store	Pinecone / pgvector	Production-grade vector search; metadata filtering for permission enforcement.
Backend	Python / FastAPI	Rapid development; strong AI ecosystem integration.
Frontend	React / Next.js	Professional UI; rapid prototyping; clean component model.
Orchestration	LangChain / custom	RAG pipeline orchestration; tool use framework.
Infrastructure	Vercel + Railway / AWS	Zero-ops deployment; demo stability; fast iteration.

13.4 Demo Objectives

- Scenario 1 — Engineering Memory: Demonstrate retrieval of an architectural decision with citations from GitHub, Slack, and Notion, including temporal confidence score.
- Scenario 2 — Onboarding Acceleration: Show a new employee query being answered with institutional context that no single tool could have surfaced alone.
- Scenario 3 — Executive Decision Context: Retrieve the reasoning behind a strategic business decision from email, Slack, and document sources.
- Scenario 4 — Knowledge Decay Alert: Surface a query result where the decay detection engine flags potentially stale information and recommends validation.

14. HACKATHON SUCCESS STRATEGY

The OpenAI x Outskill AI Builders Hackathon is evaluated across five primary dimensions. This section maps Company Brain's specific design choices to each judging criterion and articulates the strategy for maximizing scores across all five.

14.1 Technical Execution

Company Brain demonstrates technical execution through four concrete architectural choices. First, the multi-source ingestion pipeline — not trivially achieved — demonstrates real engineering capability beyond prompt wrapping. Second, the permission-enforcement architecture requires

careful design of metadata schemas, storage, and query-time enforcement logic. Third, the temporal decay model requires original algorithmic development. Fourth, the citation extraction pipeline ensures that every LLM response is grounded in verifiable source material, demonstrating sophisticated RAG implementation rather than naive generation.

Strategy: Prioritize code quality, system robustness, and architectural clarity. Prepare architecture diagrams and be ready to walk judges through the technical design in detail.

14.2 Usefulness

The usefulness case for Company Brain is grounded in quantified enterprise pain: 1.8 hours per day per knowledge worker lost to information search, \$18M+ in annual productivity loss per 500-person organization, 30–85% institutional knowledge loss per departing employee. The system directly addresses these documented, quantified pain points with measurable capability.

Strategy: Lead with the business case. Frame every demo scenario in terms of real enterprise pain and the specific value Company Brain delivers. Use concrete, relatable examples that judges can map to their own organizational experience.

14.3 Creativity and Originality

Company Brain's originality lies in its categorical innovation: the first purpose-built institutional memory infrastructure layer. It is not a better enterprise search. It is not a smarter chatbot. It is a new category of enterprise software — memory infrastructure — that addresses a structural problem no existing system is designed to solve. The temporal intelligence capability, in particular, represents a genuinely novel approach to enterprise retrieval that no existing commercial system implements.

Strategy: Name the category. "Institutional Memory Infrastructure" is a new product category, not an iteration on existing ones. Judges reward categorical thinking. Make clear that Company Brain is not a better version of Glean or Notion AI — it is a fundamentally different type of system.

14.4 OpenAI / Codex Integration

Company Brain integrates OpenAI technology at multiple layers of the stack. GPT-4o serves as the synthesis and reasoning engine, responsible for generating citation-backed institutional memory responses. text-embedding-3-large provides the semantic foundation for all retrieval operations. The OpenAI function-calling API is used to enable structured citation extraction and temporal confidence scoring. The Assistants API may be leveraged for multi-turn memory query sessions.

Strategy: Document all OpenAI API integrations explicitly. Demonstrate the specific value that OpenAI's models — and not generic alternatives — contribute to the system's capabilities.

Highlight the use of advanced API features (function calling, streaming, embeddings) to demonstrate depth of integration.

14.5 Presentation Clarity

The Company Brain narrative is structured around a single, powerful insight: organizations don't have a data problem, they have a memory problem. This framing is immediately intuitive to any experienced enterprise professional and creates a clear logical path from problem to solution to differentiation to value.

Strategy: Open with the insight, not with the features. Demonstrate live with real data. Close with the vision — memory infrastructure as the new substrate for AI-native organizations. Keep the demo narrative tight and the transitions smooth. Practice the demo until it is flawless.

15. CONCLUSION

Phase 0 of Company Brain establishes the strategic and definitional foundation for a genuinely important enterprise AI system. The problem is real, quantified, and universal. The solution is architecturally credible, technically differentiated, and market-ready. The timing is optimal — the convergence of enterprise AI adoption, distributed work normalization, and LLM capability advancement creates a precise window of opportunity for an institutional memory infrastructure layer.

Company Brain is not built to win a hackathon. It is built to solve a problem that costs enterprises billions of dollars annually and that no existing system is designed to address. The hackathon is the proving ground for an idea whose time has come.

The five strategic differentiators — temporal intelligence, permission-aware retrieval, organizational reasoning reconstruction, knowledge decay detection, and the institutional context graph — represent a coherent, mutually reinforcing architecture that creates a defensible product position in an emerging category that Company Brain is uniquely positioned to define.

***Software changed how companies operate.
Cloud changed where software runs.
AI will change who does the work.***

Company Brain ensures organizations never lose the intelligence that powers that work.

Prepared by Krishil Agrawal | OpenAI x Outskill AI Builders Hackathon | Phase 0 — Discovery & Strategic Definition